ACADEMIC DATA ANALYTICS PROJECT

Walmart Retail Data Analysis

A comprehensive exploration of **3,203 transactions** spanning four years of U.S. retail data — uncovering patterns in sales, profit, customer behavior, and product performance.

Tools Used	Period	Published
Python · Pandas · Scikit-learn · Matplotlib · Seaborn	2011 – 2014 · United States	April 2026

Presentation Outline

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Dataset Overview

3,203 transactions, 12 variables, 4 years

02

Temporal Analysis

Revenue growth and seasonal patterns

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Category Performance

Product groups, profit distribution

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Sales concentration across states

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DATASET OVERVIEW

3,203 Transactions. 12 Variables. 4 Years.

3,203

Total Transactions

Clean records with no missing values or duplicates

12

Variables

Sales, Profit, Quantity, Category, State, Customer & more

4

Years of Data

2011 through 2014 across the United States

3

Engineered Features

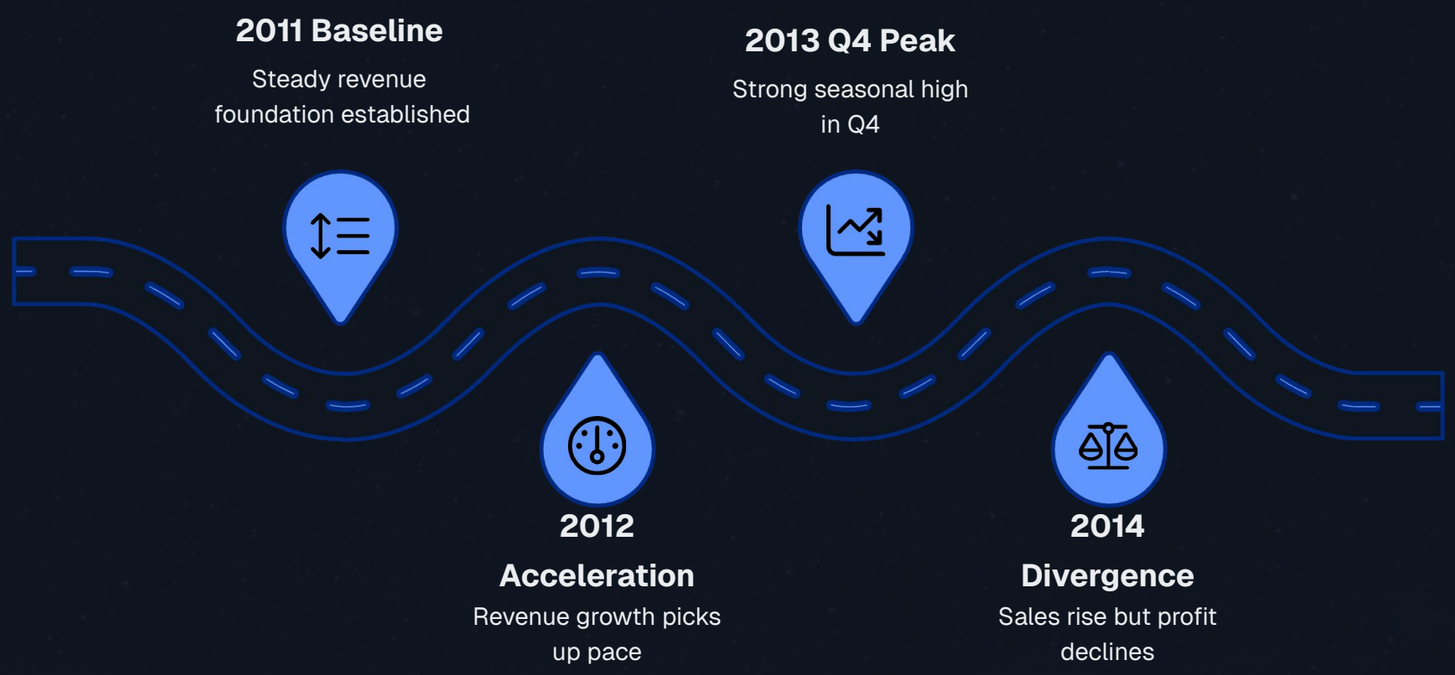
Month, Year, and Unit Price columns added

Data quality was pristine from the start — no missing values, no duplicates. Feature engineering added Month, Year, and Unit Price columns to enrich the analysis.

TEMPORAL ANALYSIS

Revenue Grew — But Profit Told a Different Story

Sales showed **consistent growth from 2011 to 2014**, with strong seasonal peaks every Q4 driven by holiday demand. However, high-revenue periods did not always translate into high profit — a critical divergence that signals hidden cost and pricing inefficiencies beneath the surface.



Q4 Holiday Effect

Recurring seasonal peaks every fourth quarter across all four years

Profit Volatility

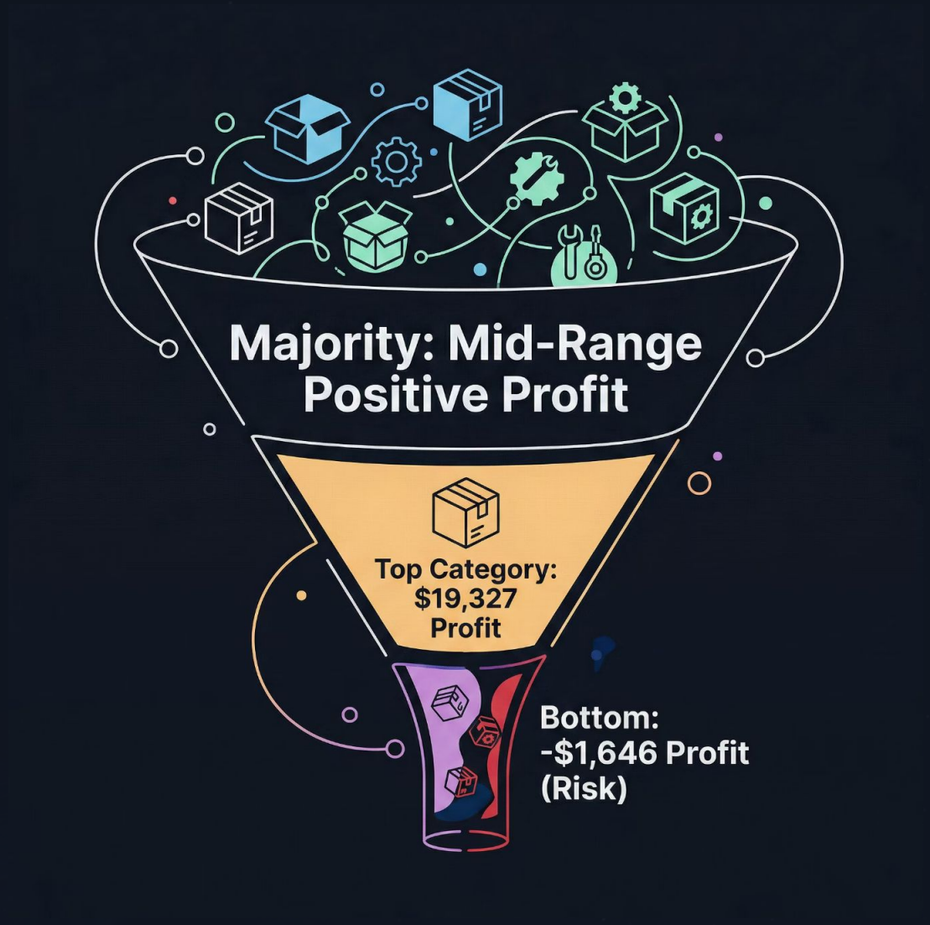
High-sales periods frequently masked margin compression and cost issues

Growth Trend

Consistent upward trajectory in revenue from 2011 through 2014

CATEGORY PERFORMANCE

17 Categories. Wide Gap Between Winners and Losers.



Across 17 product categories, performance varies dramatically. A **small minority of categories drive the vast majority of profit**, while several categories operate at a loss — representing a direct business risk.

⚠ Category-level review is essential for margin improvement. Loss-making categories must be repriced or discontinued.

🏆 **Top Category**

$19,327 in profit

⚠️ **Worst Category**

–$1,646 in profit

GEOGRAPHIC ANALYSIS

Sales Are Not Evenly Spread Across the Country

A small number of states generate a **disproportionate share of total revenue**. Regional factors — including population density, economic activity, and market penetration — explain much of this concentration.

What the Data Reveals

Revenue Concentration

A handful of states dominate total sales figures

Regional Drivers

Population density, economic activity, and market penetration are key factors

Growth Opportunity

Underperforming states represent significant untapped potential

Geographic strategy should be data-driven — targeting expansion in high-potential, currently underserved states.

CUSTOMER & PRODUCT INTELLIGENCE

686 Customers. A Few Drive Most of the Value.

686

Total Customers

Across all four years of transaction data

2.35

Avg. Order Frequency

Orders per customer over the full period

$14,345

Top Customer

Raymond Buch — highest total purchase value

$6,719

Top Product Profit

Canon imageCLASS 2200 Copier — #1 by profit

Top performing products are predominantly **office machines and technology** — a consistent signal that this category deserves prioritized investment and inventory planning.

PREDICTIVE MODELING

Can We Predict Which Products Will Win or Lose?

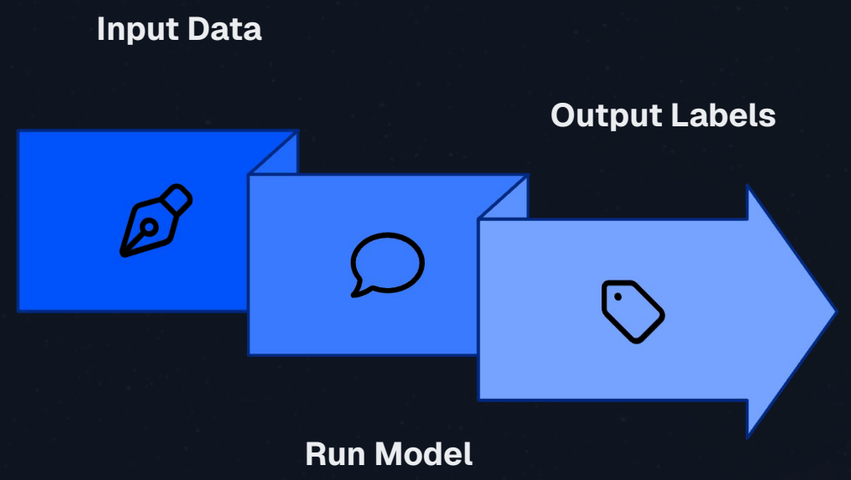
Model Summary

Goal: Classify products as **high-performing** or **loss-making** using historical transaction data.

Method: **Linear Regression** used as a classification baseline — a strong foundation for smarter inventory and restocking decisions.

Why This Matters

A 95% accuracy rate means the model can reliably distinguish between profitable and loss-making products before stocking decisions are made — transforming reactive purchasing into proactive strategy.



Accuracy	F1-Score	RMSE
95%	0.77	0.21

CLUSTERING ANALYSIS

Two Clear Product Groups Emerged from the Data

Using **K-Means clustering (2 clusters)** with PCA visualization and boxplot comparison, the data revealed a sharp divide between product groups — with pricing as the defining variable.

CLUSTER 1 - HIGH PROFIT

Median Unit Price:
~$450

Total Sales:
~$4,000

HIGH PROFITABILITY

CLUSTER 2 - LOSS MAKING

Median Unit Price:
<$100

Near-Zero/Negative Profit

LOW SALES VALUE

PRICING IS THE #1 DRIVER SEPARATING THE TWO CLUSTERS, NOT VOLUME

High Profit Cluster

Median unit price **~$450** · Total sales **~$4,000** · Strong positive margins

Loss-Making Cluster

Unit prices **below $100** · Near-zero or negative profit · High volume, low value

Key Insight: Pricing — not volume — is the #1 structural driver of profitability. High-volume, low-price products consistently underperform.

STRATEGIC RECOMMENDATIONS

What Should the Business Do Now?

Discontinue or Reprice Loss-Makers

Products identified in the loss-making cluster should be repriced above the $100 threshold or removed from inventory to protect margins.

Prioritize High-Margin Tech & Office Equipment

Stock planning should favor the high-profit cluster — particularly office machines and technology products that consistently deliver strong returns.

Align Inventory with Q4 Seasonal Peaks

Historical data confirms recurring Q4 demand surges. Inventory increases should be pre-positioned ahead of the holiday season each year.

Deploy the Predictive Model for Restocking

Use the 95%-accurate classification model to guide future purchasing decisions — shifting from reactive to data-driven inventory management.

CONCLUSION

Data Confirms: A Smarter Product Strategy Is Possible

4-Year Summary

$108,418 in total profit — driven by a concentrated set of high-performing products and categories.

The structural difference between profit and loss is **pricing**, not volume. Products priced above ~$450 consistently outperform those below $100.

The predictive model achieved **95% accuracy** in classifying product performance — ready for deployment.

Key Takeaways

1 Profit is concentrated

A small set of products and categories generate the majority of $108,418 in 4-year profit

2 Pricing is the lever

Unit price — not sales volume — is the #1 structural driver separating winners from losers

3 Model is deployment-ready

95% accuracy provides a reliable foundation for smarter restocking and inventory decisions

Next Steps: Richer datasets, advanced models (Random Forest, XGBoost), and deeper customer segmentation to unlock further value.